Veera Venkata Sai Swaroop Grandhi

Data Engineer

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Highly skilled Data Engineer with around 5 years of experience in developing robust data architectures and implementing end-to-end data processing workflows. Excited to contribute my expertise in SQL, Python, and cloud Technologies.

PROFILE SUMMARY:

- Around **5** years of experience in **Data engineering** encompassing Requirements **Analysis, Design Specification,** and **Testing** in both **Waterfall** and **Agile** methodologies.
- Hands-on experience with **Amazon EC2**, **Amazon S3**, **Amazon RDS**, **VPC**, **IAM, Amazon Elastic Load Balancing, Auto Scaling, Cloud Watch, SNS, SES, SQS, Lambda, EMR** other services of the **AWS family.**
- Worked with **csv, Avro, Parquet** data to load into **Data frames** and do the analysis. Hands on experience on **Google Cloud Platform (GCP)** in all the big data products **BigQuery, Cloud Data Proc, Google Cloud Storage, Composer (Air Flow as a service).**
- Proficiency with **Scala**, **Apache HBase, Hive, Pig, Sqoop, Zookeeper, Spark, Spark SQL, Spark Streaming, Kinesis, Airflow, Yarn, and Hadoop (HDFS, MapReduce).** Implemented **Big Data solutions** using **Hadoop** technology stack, including **Pyspark**, **Hive**, **Sqoop**, **Avro** and **Thrift**.
- Working knowledge of **HDFS**, **Kafka**, **Map Reduce**, **Spark**, **PIG**, **HIVE**, **Sqoop**, **HBase**, **Flume**, and **Apache Zookeeper** as tools for designing and deploying end-to-end big data ecosystems.
- Hands-on experience with **Spark**, **Data bricks**, and **Delta Lake** and Practical experience with **Python** and Apache **Airflow** to create, schedule, and monitor workflows. Experience in automating day-to-day activities by using **Windows PowerShell**.
- Experienced working in both **Agile** and **Waterfall** based development environment and participating in **Scrum** sessions.
- Practical experience with **Python** and **Apache Airflow** to create, schedule, and monitor workflows. Experienced in building **Snow Pipes,** migrating **Teradata** objects into **Snowflake** environment.
- Proficient with Container systems like **Docker** and Container orchestration like **EC2** Container Service, **Kubernetes**, worked with **Terraform**. and Experience in implementing Azure data solutions, provisioning storage account, **Azure Data Factory, SQL server, SQL Databases, SQL Data warehouse, Azure Data Bricks and Azure Cosmos DB.**
- Expert in designing Parallel jobs using various stages like **Join, Merge, Lookup, remove duplicates, Filter, Dataset, Lookup file set, Complex flat file, Modify, Aggregator, XML.**
- Hands-on experience using **Snowflake** utilities, **SnowSQL, SnowPipe, Big Data model techniques using Python / Java.**
- Expertise in building **CI/CD** on **AWS** environment using **AWS** Code **Commit, Code Build, Code Deploy** and **CodePipeline** and experience in using **AWS CloudFormation**, **API Gateway**, and **AWS Lambda** in automation and securing the infrastructure on **AWS**.
- Proficient in building **CI/CD** pipelines in **Jenkins** using pipeline syntax and **groovy** libraries and Excellent knowledge on **Hadoop** Architecture and ecosystems such as **HDFS, Hive, Pig, Sqoop, Job Tracker, Task Tracker, Name Node, Data**
- Involved in using **Jenkins**, **Selenium**, **Docker** and **Kubernetes** clusters and **power Shell scripting** for efficient data management and decentralized access and Experience working with **Azure Logic APP** Integration tool and working with Data warehouse like **Oracle, SAP, HANA.**
- Instantiated, created, and maintained **CI/CD** (continuous integration & deployment) pipelines and apply automation to environments and applications and worked on various automation tools like **GIT, Terraform, Ansible.**

TECHNICAL SKILLS:

Programming Languages:

- **Python:** Widely used for scripting, data manipulation, and developing data pipelines.
- **SQL:** Essential for querying and managing relational databases.
- **Java/Scala:** Useful for working with big data tools like Apache Spark and Hadoop.

Data Warehousing & Databases:

- **Relational Databases:** MySQL, PostgreSQL, SQL Server.
- **NoSQL Databases:** MongoDB, Cassandra, HBase.
- **Data Warehousing:** Amazon Red shift, Google BigQuery, Snowflake.

Big Data Technologies: HDFS, Hue, MapReduce, PIG, Hive, HCatalog, HBase, Sqoop, Impala, Zookeeper, Flume, Kafka, Yarn, Cloudera Manager, Kerberos, PySpark Airflow, Kafka, Snowflake Spark Components

Cloud Platforms:

- **AWS:** Services like S3 (storage), Redshift (data warehousing), Lambda (serverless computing), and EMR (Hadoop/Spark).
- **Azure:** Services like Azure Data Lake, Azure Synapse Analytics, and Azure Databricks.
- **Google Cloud Platform (GCP):** BigQuery, Cloud Dataflow, and Cloud Storage.

Visualization& ETL tools: Tableau, PowerBI, Informatica, Talend,IICS

Web/Application server: Apache Tomcat, WebLogic, WebSphere

Version controls and Tools: GIT, Maven, SBT, CBT

Data Processing Frameworks: Apache Hadoop, Apache Spark, Apache Flink, Apache Beam.

WORK EXPERIENCE:

Key Bank | Azure Data Engineer

Cleveland, Ohio, USA **July 2024 - Present**

OBJECTIVE: As an Azure Data Engineer at **Key Bank**, you will be responsible for designing, developing, and maintaining robust data solutions on Microsoft Azure cloud platforms. You will play a key role in managing data pipelines, optimizing data storage, and ensuring data quality and governance to support analytics, reporting, and business intelligence initiatives.

Key Responsibilities:

- Led the migration of an on-premises **Data Warehouse to Azure Synapse Analytics**, enhancing scalability and **reducing costs** by **30**%.
- Built a data quality framework in Synapse Notebooks and Pipelines to ensure **99% data accuracy** and **consistency** across pipelines.
- Leveraged **Azure Synapse Serverless SQL Pools** for on-demand querying of large datasets, cutting **infrastructure costs by 40%.**
- Applied data governance best practices, ensuring compliance with **GDPR/CCPA**, and maintaining audit trails and data lineage.
- Implemented **Delta Tables** to enable incremental loading and maintain versioned data, improving **data reliability by 25%.**
- Integrated data sources, including **SQL Data Warehouse, Azure Blob Storage**, and external databases, ensuring seamless reporting workflows.
- Automated data ingestion and transformation pipelines in Synapse, **reducing ETL run times by 20%.**
- Designed **data validation scripts to flag missing values**, outliers, and duplicates, enhancing data quality for analytics by 30%.
- Streamlined ETL workflows using Azure Synapse Pipelines, driving faster data processing, and **reducing manual effort**.
- Partnered with cross-functional teams to understand data requirements and delivered scalable analytics solutions.
- Developed complex SQL solutions, including views, stored procedures, and triggers, optimizing database performance.

- Used **Git and GitHub** for version control and automation of deployment pipelines, ensuring efficient collaboration and streamlined resource deployments.
- Collaborated with security teams to ensure data solutions adhered to strict banking industry regulations.
- Utilized **Azure Monitor and Log Analytics** for real-time performance monitoring and troubleshooting.
- Improved software delivery processes by integrating **GitHub** Actions into **the CI/CD framework**, accelerating releases, and ensuring high-quality deployments.
- Built Spark-Streaming applications to **process Kafka** data streams and ingest real-time data into HBase.
- Designed insightful Tableau dashboards, visualizing KPIs and providing actionable business insights.
- **Environment:** Azure Synapse Analytics, SQL, Git, GitHub, Azure Blob Storage, Azure Monitor, Azure Databricks, Delta Lake, Apache Kafka, Spark Streaming, HBase, Tableau, PySpark, Apache Spark, CI/CD, GDPR, CCPA, Docker, Jupyter Notebooks, Azure Kubernetes Service, Snowflake, Data Governance, ETL Pipelines, Azure Serverless SQL Pools, Data Validation, Real-time Data Processing, Spark SQL, Log Analytics.

Root Insurance | AWS Data Engineer

Columbus, Ohio, USA **Dec 2023 - June 2024**

OBJECTIVE: Root Insurance Company is a technology-driven auto insurance provider primarily on individual driving behaviors, aiming to provide fair and affordable insurance to good drivers. Ensuring data security across AWS platforms, including creating and maintaining IAM users and roles and creating and implementing data governance processes to mitigate security and privacy risks

Key Responsibilities:

- **Designed and implemented scalable data architectures** on AWS to support Root Insurance's data ingestion, storage, processing, and analytics needs.
- Built robust data pipelines leveraging **AWS Glue, Lambda, Kinesis, and S3 for seamless real-time** and batch data processing.
- Optimized ETL processes to integrate data from diverse sources into **AWS data lakes and data warehouses** like Amazon Redshift, improving query performance by 30%.
- Automated workflows and orchestrated data integration using **AWS Step Functions** and Data Pipeline, reducing manual intervention by 40%.
- Managed and optimized data warehousing solutions in **AWS,** ensuring data accessibility for business intelligence and analytics, enabling faster decision-making.
- Utilized **Amazon Redshift, RDS, and Aurora** to handle structured data, ensuring scalability and superior performance for insurance analytics.
- Implemented advanced data security protocols such as **encryption (AWS KMS)** and granular access controls to protect sensitive insurance data.
- Used **AWS CloudWatch and AWS CloudTrail** for real-time monitoring, **logging**, and performance optimization across the data ecosystem.
- **Developed CI/CD pipelines** using AWS CodePipeline, CodeBuild, and CodeDeploy to automate deployment and testing, **reducing rollout time by 50%**.
- Wrote complex SQL queries for data extraction, aggregation, and analysis from databases like **Amazon RDS (MySQL, PostgreSQL) and Redshift, improving data insights.**
- Automated resource management and workflows using Python with AWS SDK (Boto3), enabling custom data operations with reduced effort.
- Automated infrastructure deployment using **AWS CloudFormation** and **streamlined CI/CD practices,** ensuring consistency across deployments.
- Enabled real-time data streaming through **Amazon Kinesis,** supporting time-critical analytics and ensuring low-latency data processing.

- Developed a serverless Lambda function integrated with S3 to automate event-based **data workflows, reducing operational overhead.**

Environment: AWS, AWS Glue, AWS Lambda, Amazon Kinesis, Amazon S3, Amazon Redshift, AWS data lakes, Amazon RDS, Amazon Aurora, HIPAA, AWS KMS, Amazon Elastic ache, AWS CloudWatch, AWS Cloud, SQL, Python, AWS SDK.

Novo Nordisk (Cognizant) | Informatica Developer

Hyderabad **June 2022 – July 2023**

OBJECTIVE: Demonstrated proficiency in leveraging cloud-based and on-premises data integration platforms, with expertise in Informatica and Snowflake to design robust ETL frameworks, optimize data workflows, and empower data-driven decision-making.

Key Responsibilities:

- **Designed and optimized ETL pipelines using Informatica PowerCenter, IICS and Snowflake**, delivering high-performance data integration and seamless flow across multiple data sources, enhancing overall system efficiency.
- Led data transformation and cleansing initiatives within Informatica, ensuring data integrity and high-quality processing that drives actionable insights across business domains.
- Built and maintained **scalable Snowflake data warehouses**, improving reporting **performance by 40%** and enabling more efficient querying to support timely business decisions.
- Developed advanced data models in **Snowflake (including Databases, Schemas, Tables)**, ensuring optimization for large data sets and significantly improving ETL process speeds.
- Integrated diverse data sources into Snowflake via **Informatica, guaranteeing error-free, on-time** loading and transformation of critical business data across environments.
- Automated key ETL workflows in Informatica, **reducing manual intervention by 30% and accelerating** data extraction, transformation, and loading processes for rapid insights.
- Utilized Snowflake's advanced features (e.g., Time Travel, Zero-Copy Cloning) for robust data management and disaster recovery, ensuring minimal downtime and maximum system availability.
- Optimized performance of **Informatica PowerCenter and Snowflake environments**, reducing query times and operational costs, boosting overall platform performance.
- **Validated data integrity across complex ETL pipelines,** using automated testing frameworks to ensure consistent and reliable results, **reducing errors by 20%.**
- Built interactive **dashboards in Tableau and Power BI** that connected directly to live Snowflake and Informatica data, providing actionable, **real-time insights to business teams**.
- **Environment:**

  Informatica PowerCenter, Snowflake, SQL, Python.

Bayer | Data Engineer

Hyderabad **June 2019 - May 2022**

OBJECTIVE: Bayer is a company with core competencies in the life science fields of healthcare and agriculture. Designed and built data warehouses, core data models, and ETL/ELT processes and Prototyping analytics views and building systems to answer business questions Creating data-driven tools, templates, packages, visualizations, and dashboards.

Key Responsibilities:

- **Built and optimized scalable data pipelines using Python, PySpark, Hive SQL,** and **Presto**, improving data extraction and transformation from relational and non-relational databases (e.g., Cassandra, HDFS) by 30% in speed and efficiency.
- **Designed and implemented ETL pipelines** with **Spark and Hive**, reducing data ingestion time by 25% while processing data from multiple sources more reliably.
- **Led data transformation and cleansing efforts** using **SQL, Python, and PySpark,** ensuring 98% data accuracy and improving consistency across large-scale datasets.

- Expertly utilized **Hive SQL, Presto SQL, and Spark SQL** to design and optimize **ETL processes**, enhancing data processing **throughput by 20%.**
- **Managed ETL solutions** and automated operational workflows, **implementing SQL Server Integration Services (SSIS)** for data validation, reducing manual intervention by 40% and accelerating data processing timelines.
- **Implemented containerized solutions** with **Docker** and **Kubernetes,** automating CI/CD pipelines and improving deployment **efficiency by 35%** for cloud-based custom application images.
- **Designed and configured back-end applications** and databases, managing large datasets **with Pandas and SQL**, and utilizing **AWS Terraform** to reduce manual infrastructure provisioning time by **50%**.
- **Developed and maintained interactive dashboards in Tableau**, connecting to data sources like **BigQuery and Presto SQL**, increasing data-driven decision-making speed by 30% for business stakeholders.
- **Proactively monitored server performance with Nagios**, CloudWatch, and **the ELK Stack (Elasticsearch, Kibana),** reducing downtime and improving server stability by 25%.

Environment: python, Pyspark, HiveSQL, Presto, HDFS, Presto SQL, Docker, Kubernetes, Linux, Git, Jenkins, Pandas, AWS Terraform, Nagios, Cloud watch, Elastic search, Kibana.

EDUCATION:

University of Cincinnati, Masters in Information Technology, USA from 2023 – 2024.